Speaking Script

Slide 1 — Title

Hey everyone, my name is Ashton Smith and this is my midterm project for ITAI 1378, Computer Vision and AI.

My project is called **Smart Trash Sorting Using Computer Vision.**

The goal of this project is to use artificial intelligence to automatically identify trash items and help people dispose of them correctly.

Slide 2 — The Problem

The problem I am focusing on is incorrect trash sorting.

A lot of recyclable materials end up in the wrong trash bins.

This creates more landfill waste and also makes recycling more expensive for cities and waste management facilities.

The people who care about this problem include recycling centers, environmental organizations, and local governments.

Slide 3 — The Solution

The solution I am proposing is a computer vision system that can identify trash items automatically.

A camera would capture an image of the item, and the AI model would analyze it.

The system would then classify the item and suggest whether it belongs in a recycling bin or a regular trash bin.

In simple terms, this system helps guide people to throw away items the correct way.

Slide 4 — Technical Approach

This project uses computer vision, which allows computers to analyze and understand images.

The model used is YOLOv8, which is a popular object detection model that is known for being fast and accurate.

The system would be built using PyTorch along with Python and computer vision libraries like OpenCV.

YOLOv8 is a good choice because it performs very well in real time object detection tasks.

Slide 5 — Data Plan

To train the model, a dataset of labeled images is needed.

These images could come from public datasets on platforms like Kaggle or Roboflow.

The dataset would include common trash items such as plastic bottles, aluminum cans, paper, and general waste.

By training on hundreds or thousands of labeled images, the model learns how to recognize these different types of objects.

Slide 6 — System Diagram

This slide shows the basic flow of how the system would work.

First, the system captures an image using a camera.

Next, the YOLOv8 model processes the image and detects the object.

Then the system classifies the item into the correct category.

Finally, the system outputs the recommended bin for the item.

So overall the system goes from image acquisition, to model inference, to output visualization.

Slide 7 — Success Metrics

To measure the success of this system, we look at two main metrics.

The first is accuracy, which measures how often the model correctly classifies the trash item.

The goal would be to achieve over 90 percent accuracy.

The second metric is speed, meaning how quickly the model can process an image.

Ideally the system should classify an item in less than one second.

Slide 8 — Week by Week Plan

The project timeline starts with collecting a dataset and setting up the development environment.

Next, the model would be trained using the labeled images.

After training, the model would be tested and improved to increase accuracy.

Then a simple demo or visualization of the system would be created.

Finally, testing and documentation would be completed before presenting the project.

Slide 9 — Challenges and Backup Plans

One challenge could be not having enough labeled training data.

If that happens, a larger dataset from Roboflow could be used.

Another challenge could be low model accuracy.

If that occurs, techniques like data augmentation or tuning model parameters could help improve performance.

Slide 10 — Resources Needed

The main computing platform for this project would likely be Google Colab or Kaggle.

The frameworks used would include PyTorch and the Ultralytics YOLO library.

The estimated cost for this project is very low, likely between zero and ten dollars depending on compute usage.